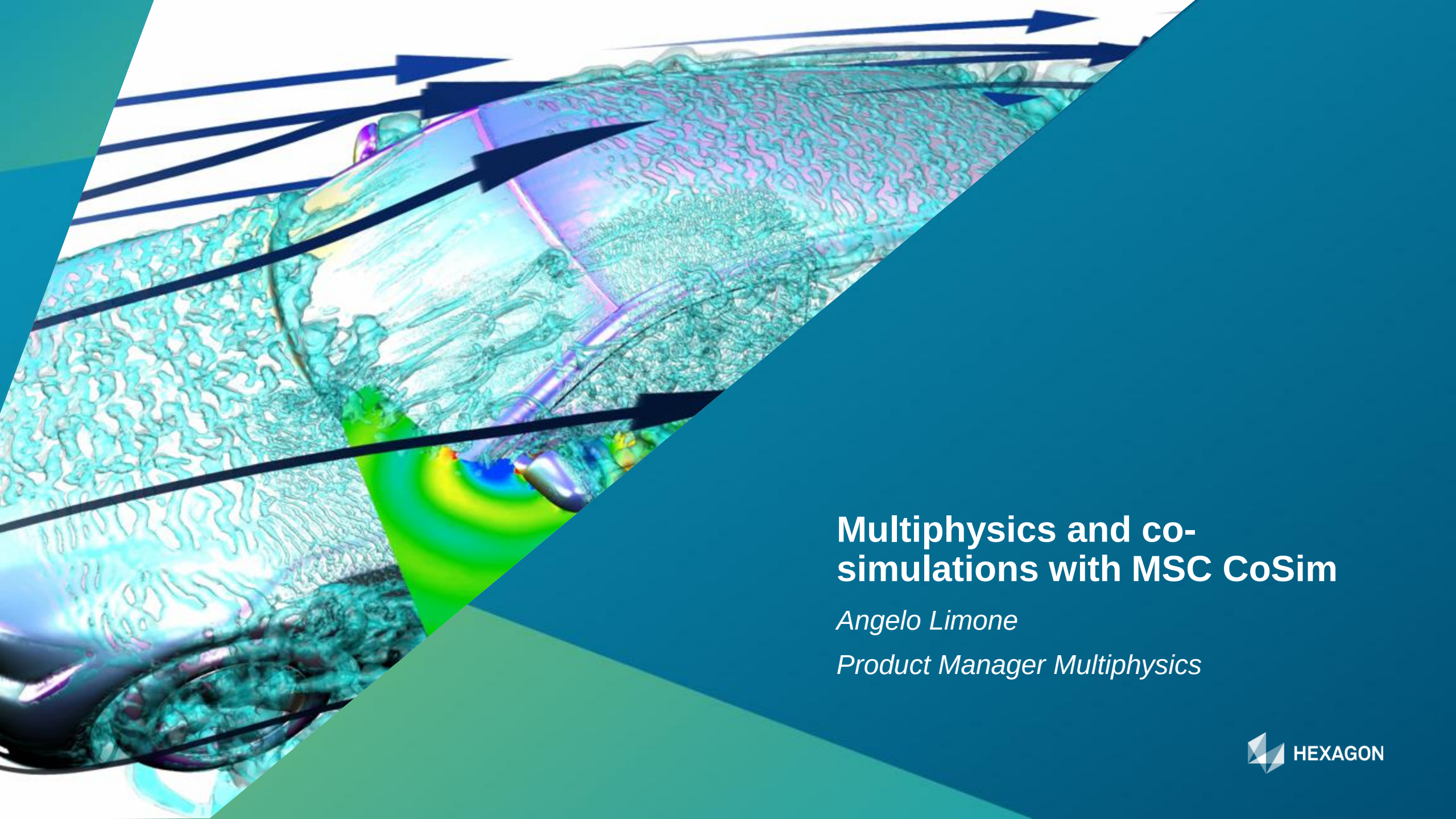




Multiphysics and co-simulations with MSC CoSim

Angelo Limone

Product Manager Multiphysics

Agenda

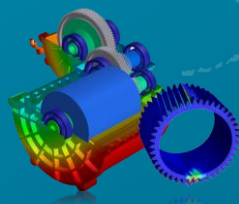
-  **1 Co-simulation and Multiphysics: What is MSC CoSim?**
-  **2 MSC CoSim 2022: Current version and capability**
-  **3 Short gallery of cases**
-  **4 MSC CoSim 2022.1**

What is MSC CoSim?

Multiphysics via co-simulations

Our Mission

Gear /
Transmission Design



Romax

Rx

ADAS /
Virtual Test Driving

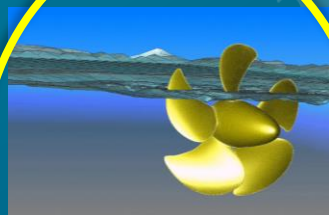


Virtual Test Drive

VTD

Common CAE Platform

- Modeler
- Format Conversion
- Generative Design (Topology Optimization)



Cradle

Cr^{FL} CrST

CFD / DEM

Multi-body
Dynamics



Adams

Ad

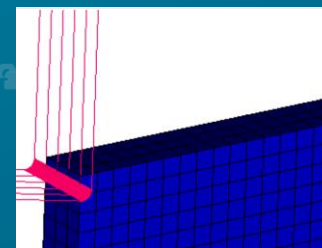
Elastic deformation /
Vibration / Yield



MSC Nastran

Na

Superelastic /
Ductile Fracture



Marc

Ma

Impact / Brittle Fracture / FSI



Dytran

Dy

Fatigue / Fracture



CAEfatigue

Cf

Co-simulation

MSC Cosim

Co^s

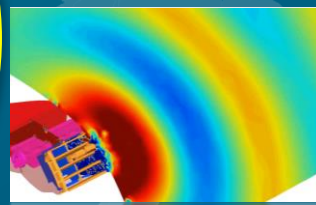
MSC Apex

Ap

ODYSSEE
TRANSFORMING DATA INTO INTELLIGENCE

CAE
ODS

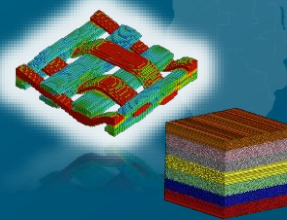
ML / Optimization



Actran

Ac

Acoustic / NVH



Digimat

Di

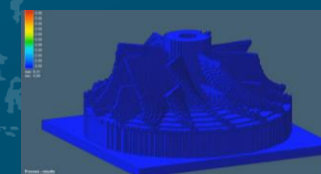
Material Modeling



FTI FormingSuite

FS

Formability



Simufact

Sf

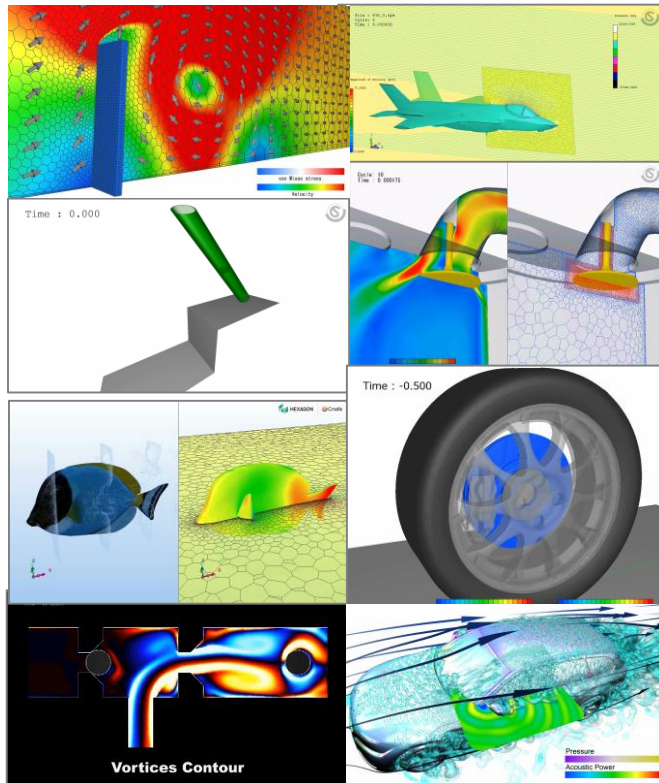
Production Technology

- Metal Forming
- Welding
- Additive Manufacturing (3D Printing)

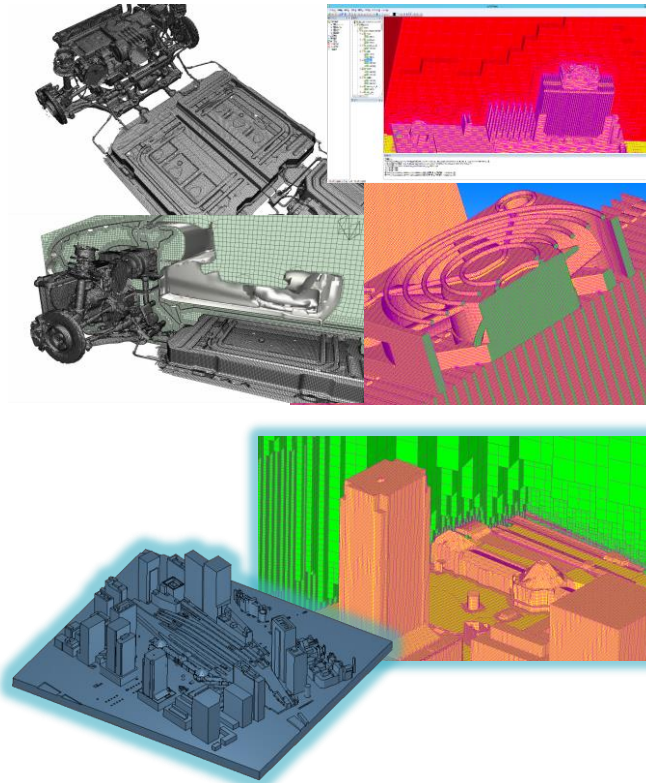
Hexagon's Cradle CFD toolset

3 main core competences

Multiphysics: co-simulation via MSC One



Productivity: Stability & Speed of mesher & solver



Visualization: photorealistic & immersive



MSC CoSim 2022

Adams

- Multi-body dynamics simulation tool
- Dynamics of moving parts
- Distribution of loads and forces throughout mechanical systems

Marc

- General-purpose nonlinear finite element analysis solution
- Static, dynamic and multi-physics loading scenarios

MSC Nastran

- Well-known solver for linear and non-linear analyses under static and dynamic loading conditions

Cradle CFD

- **scFLOW**, CFD for thermo-fluid analysis with polyhedral mesh
- **scSTREAM**, CFD for thermo-fluid analysis with structured mesh



CoSim Coupling Available @ CoSim 2022

MSC Nastran



Adams



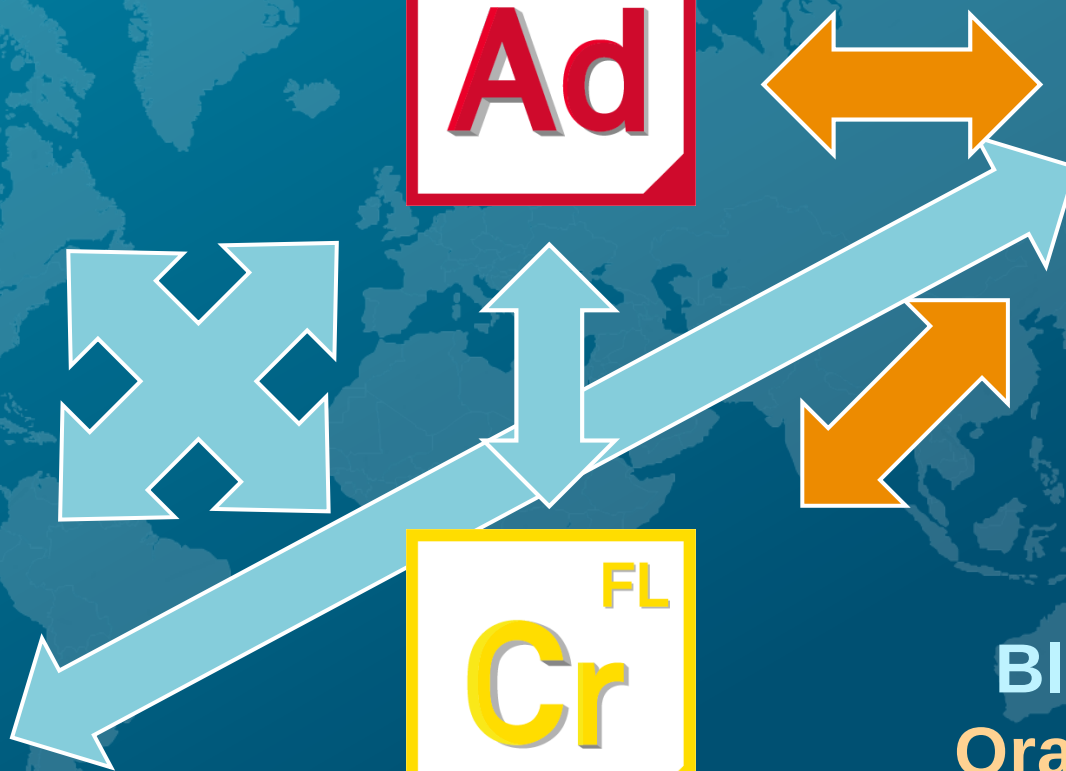
Marc



scSTREAM



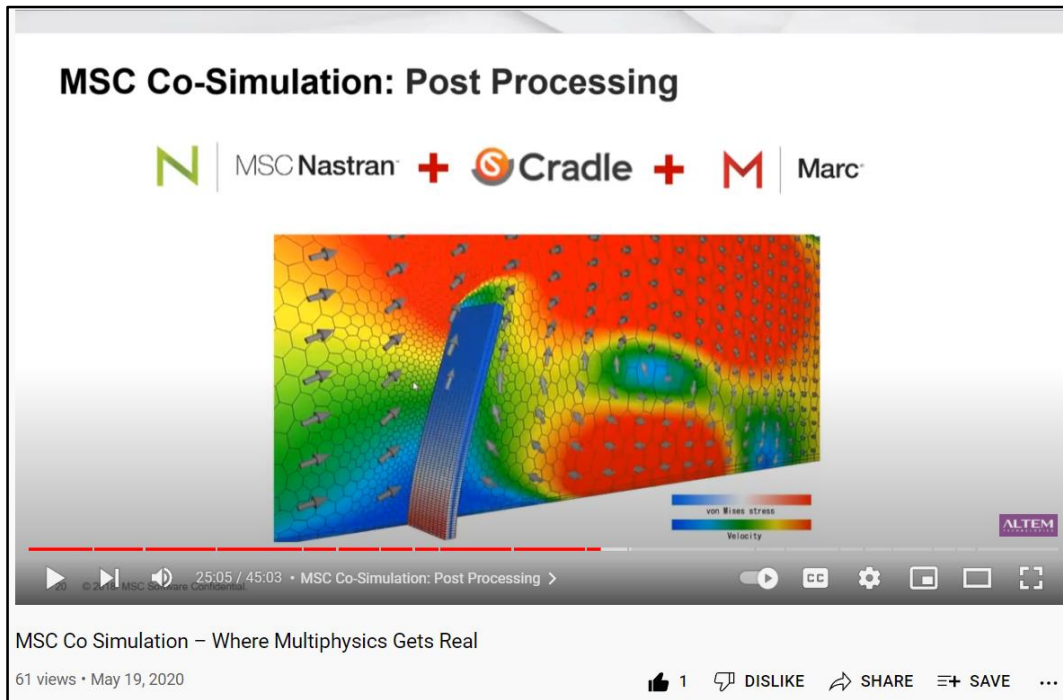
scFLOW



Blue : 2 way
Orange : 2 or 3 way

More Information

Co-simulation using MSC CoSim



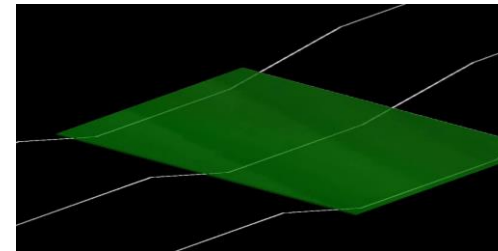
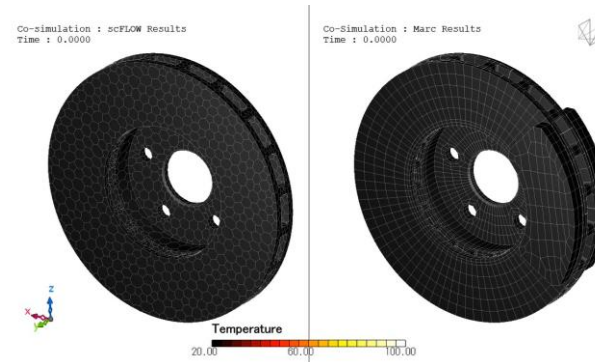
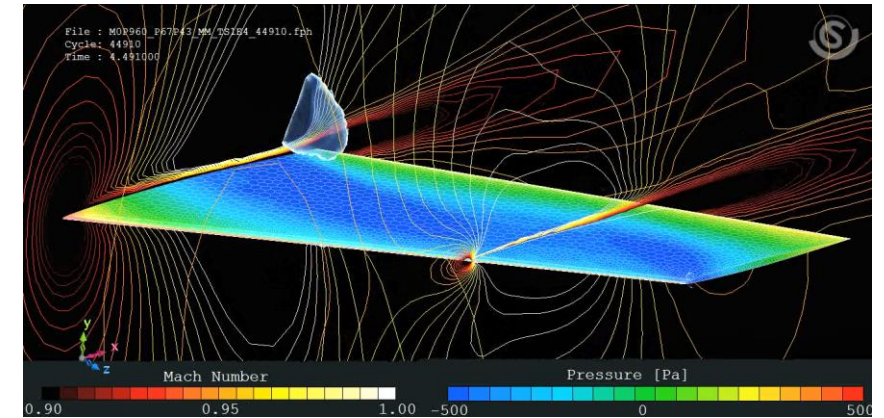
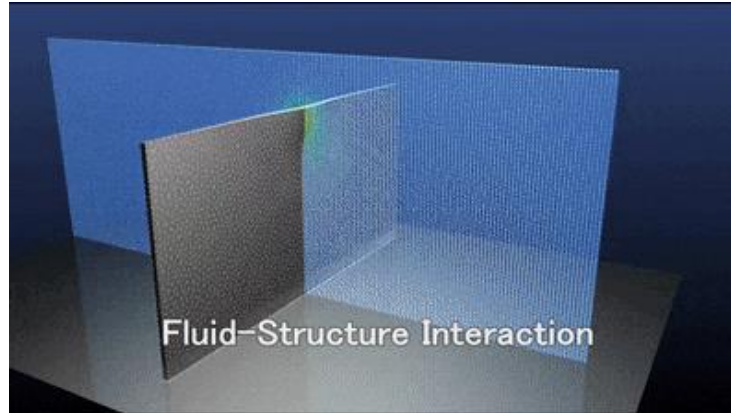
45 min video on <https://www.youtube.com/watch?v=6-P61Qbluf0&t=388s>



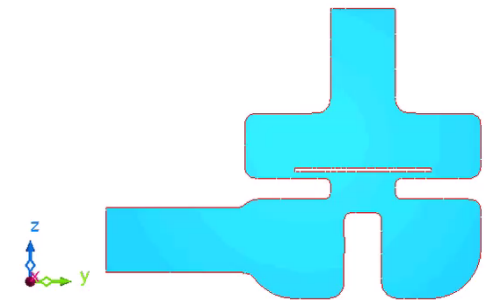
<https://hexagon.com/products/msc-cosim>

Exemplary applications

- Fluid-structure interaction
- Aeroelasticity
- Waterway drive tests
- Offshore floating platforms & marine
- Thermal stress and fatigue on solid structures
- Airbags
- Door slamming
- Thermal stress in electronics
- Suspension dynamics
- Wipers dynamics
- Rubber valves
- Fuel tank sloshing



File : flexible_membrane_scflow_0.fph
Cycle: 0



MSC CoSim 2022

April 2022

Enhancement of Adams – Marc coupling

Cut-back support

- **What:**

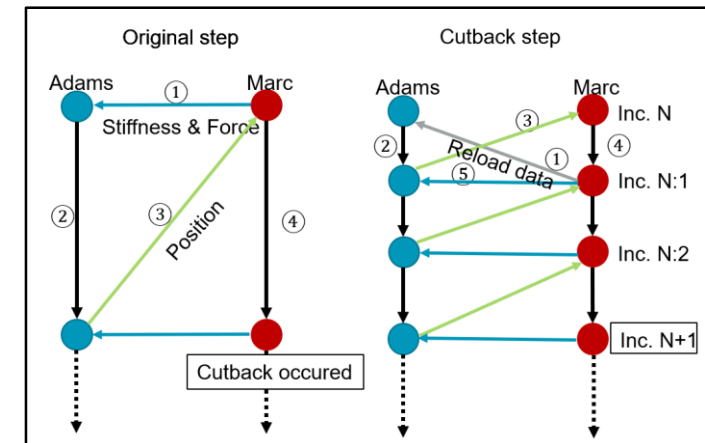
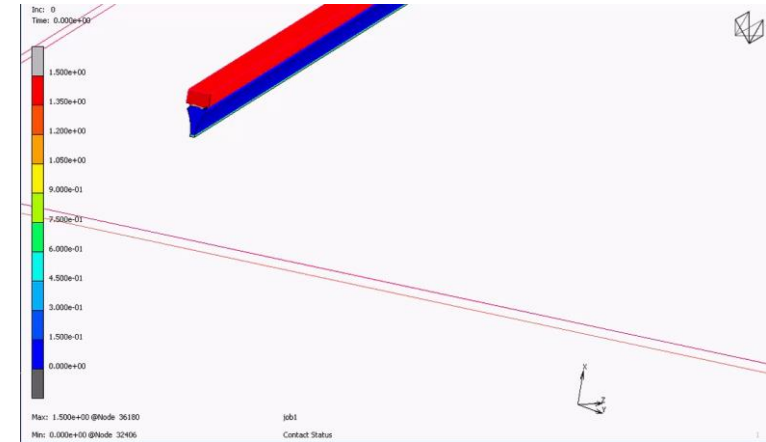
- Revised co-simulation scheme uses a more advanced data exchange orchestration
 - During cutback, Marc uses the updated position calculated by Adams
 - New scheme better accounts for non-linearities and prevents instabilities

- **Benefit:**

- Improved stability and accuracy of cases with non-linear dynamics

- **Target:**

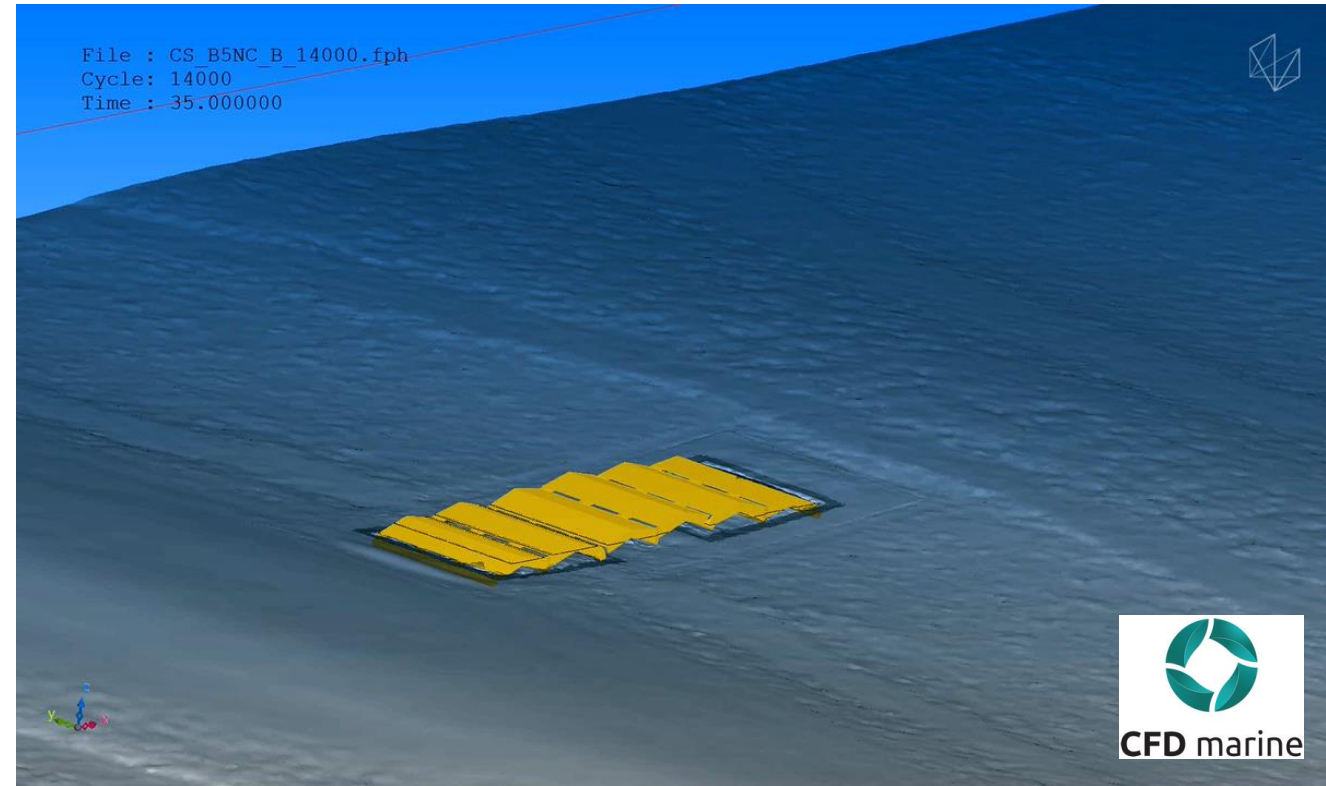
- Wipers dynamics
- Door slamming
- Suspensions dynamics



Enhancement of Adams – scFLOW coupling

Virtual Mass

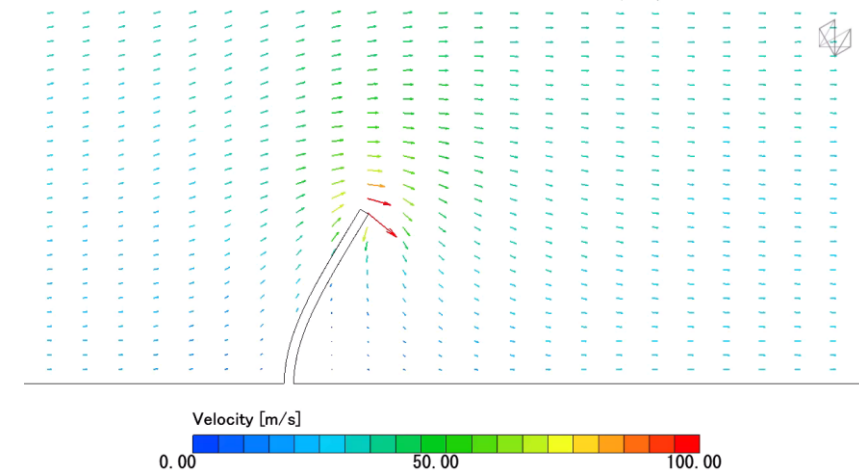
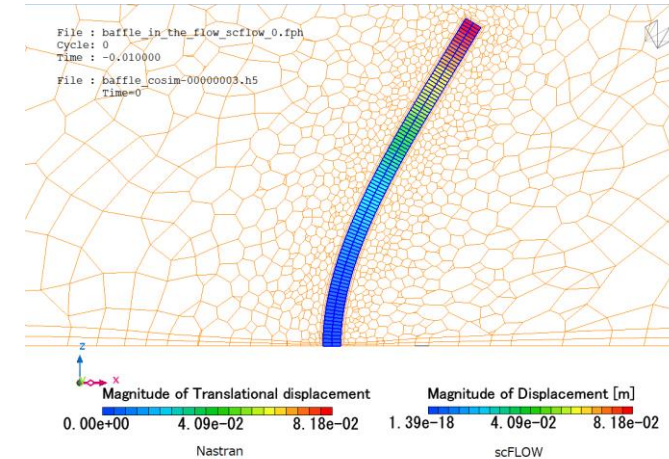
- **What:**
 - Introduction of Virtual Mass, which effectively accounts for the additional inertial effect caused by surrounding fluids on the dynamics of immersed bodies
- **Benefit:**
 - Improved stability of simulations cases where solid bodies are immersed or partially immersed
 - More user-friendly setups thanks to reduced parameter tuning
- **Target:**
 - Floating structures:
 - Offshore wind turbines
 - Telegraphic cables in ocean
 - Marine applications



Enhancement of MSC Nastran – scFLOW coupling

Initial static analysis

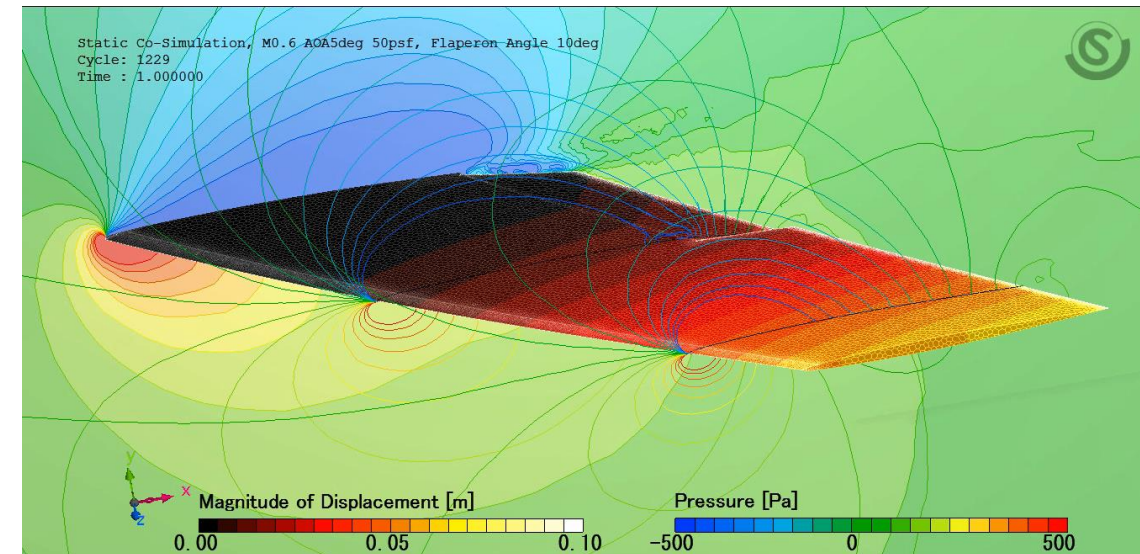
- **What:**
 - The initial static analysis allows users to run transient scFLOW with static MSC Nastran for inexpensive preliminary computations
 - MSC Nastran completes the static analysis before invoking CoSim, then scFLOW accounts for the displacement just before running in transient or steady state
- **Benefit:**
 - Significantly reduced time-to-solution of cases involving exchange of pressure, viscous stress and displacement
- **Target:**
 - Aeroelasticity of aerostructures
 - Wind turbine blades
 - Hydrofoils of sailboats
 - Fuel tank sloshing



Enhancement of MSC Nastran – scFLOW coupling

Static FSI analysis by CoSim iteration

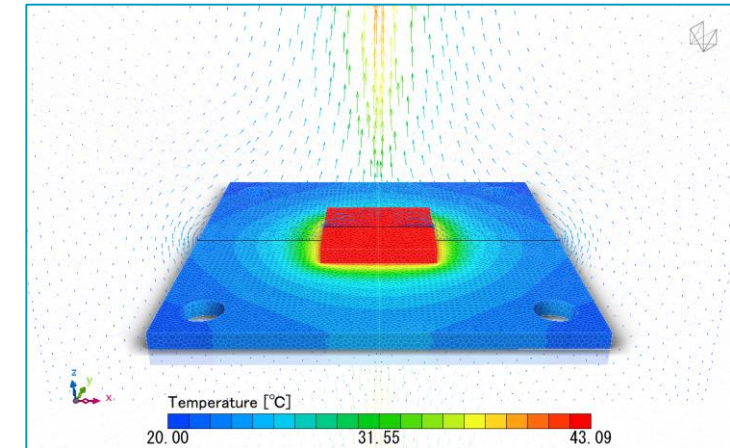
- **What:**
 - CoSim iteration algorithm can now be applied to static Fluid-Structure-Interaction analyses
 - MSC Nastran performs static non-linear analysis and scFLOW performs steady analysis, exchanging displacement and pressure/force with each other, and repeating this until the fluctuation of exchanging values becomes small
- **Benefit:**
 - Steady-state solutions of Fluid-Structure Interaction analyses can now be obtained without having to perform long transient analyses
- **Target:**
 - Aeroelasticity of aerostructures
 - General FSI cases
 - Turbine blades and casing design



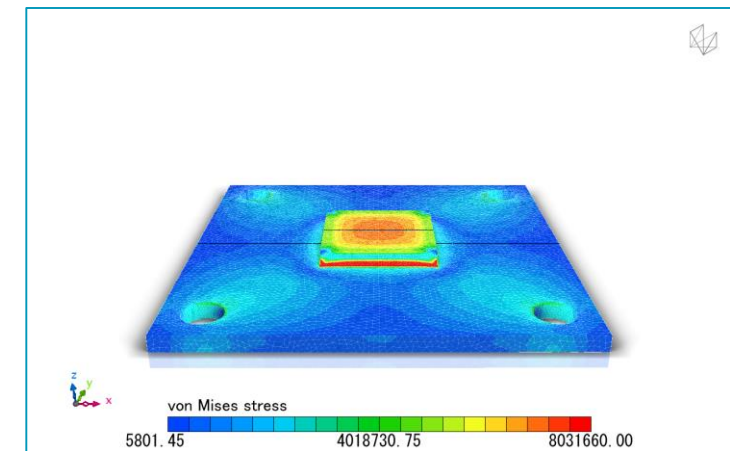
Enhancement of MSC Nastran – scFLOW coupling

1-way thermal load analysis

- **What:**
 - The temperature field computed by scFLOW can now be mapped onto Nastran in transient cases
- **Benefit:**
 - Beneficial for thermal stress in solids in transient
 - Intermediate feature for ongoing enhancements of hypersonics capabilities
- **Target:**
 - Thermal stress analysis of solid structures
 - E.g. thermally-induced fatigue on mixing pipes (T-joints)
 - Hypersonic applications such as re-entry vehicles
- **Notes:**
 - Rankin temperature is not supported



Temperature and flow analysis by scFLOW

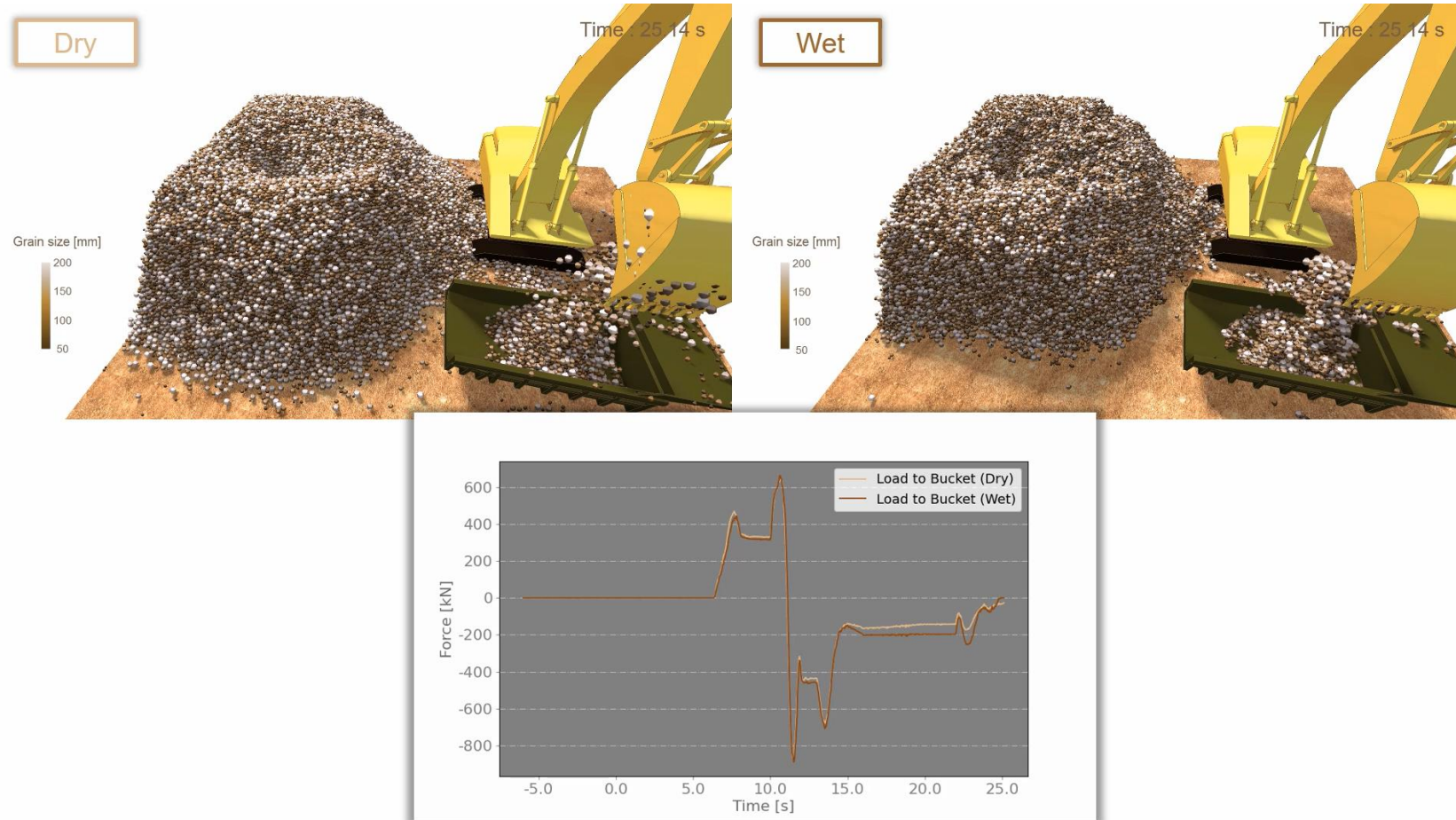


Thermal stress analysis by Nastran

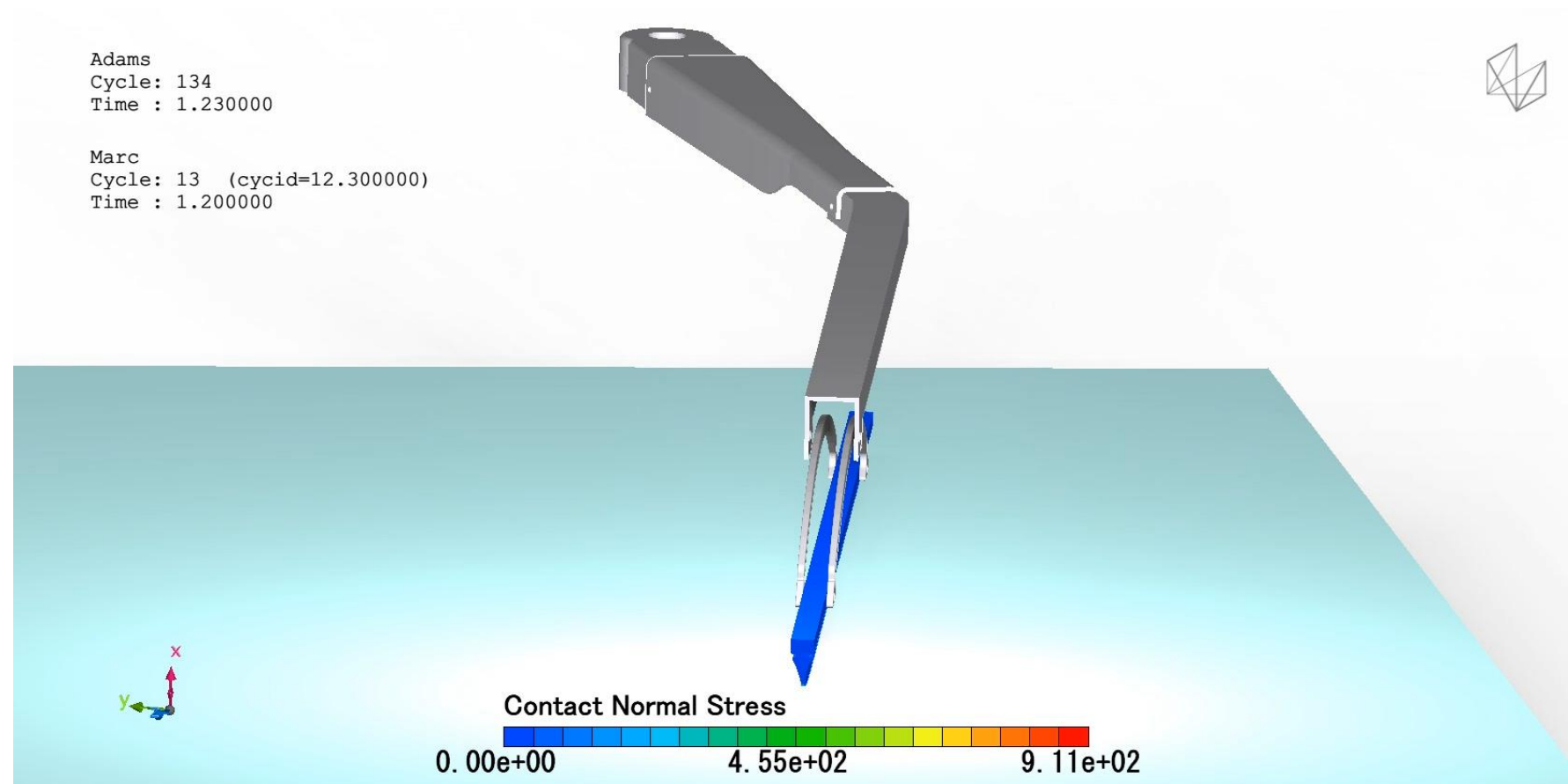
Short gallery of cases

Beyond classic CFD

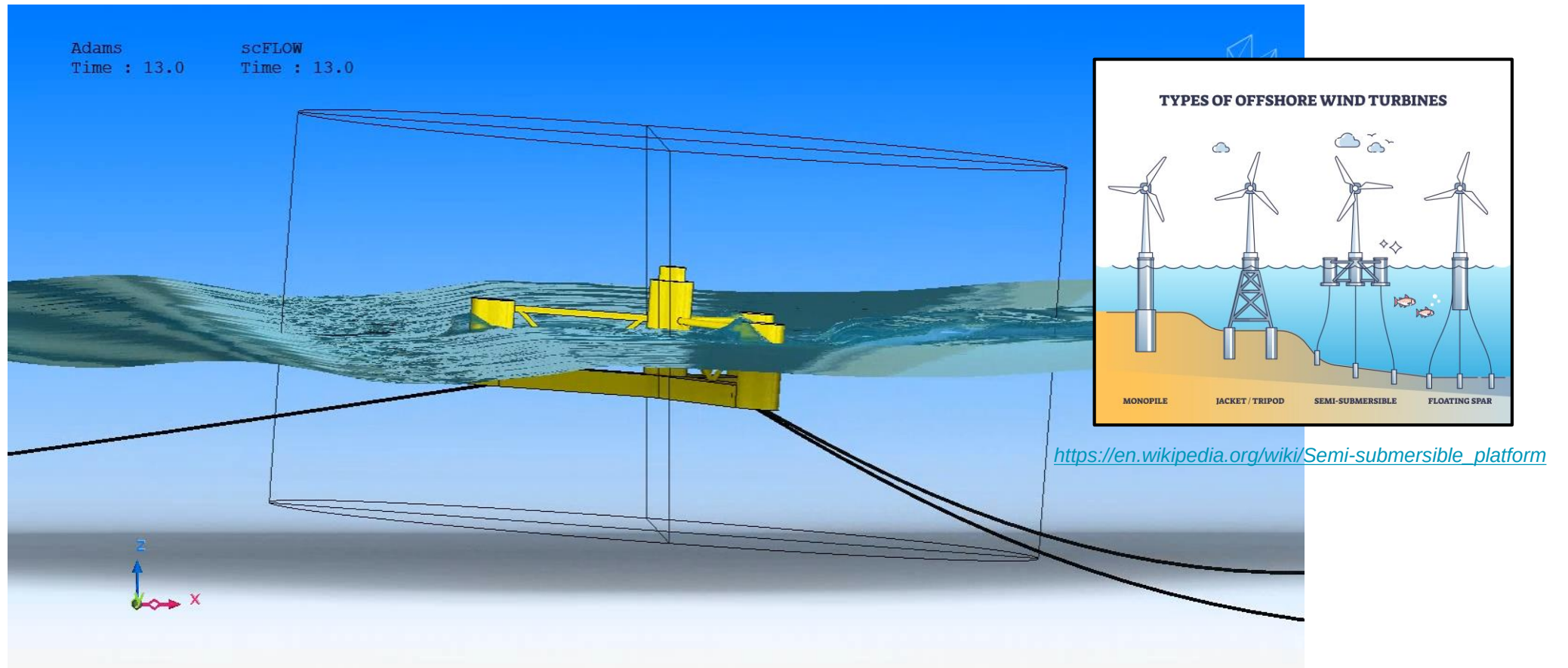
scFLOW – Adams interaction via MSC CoSim



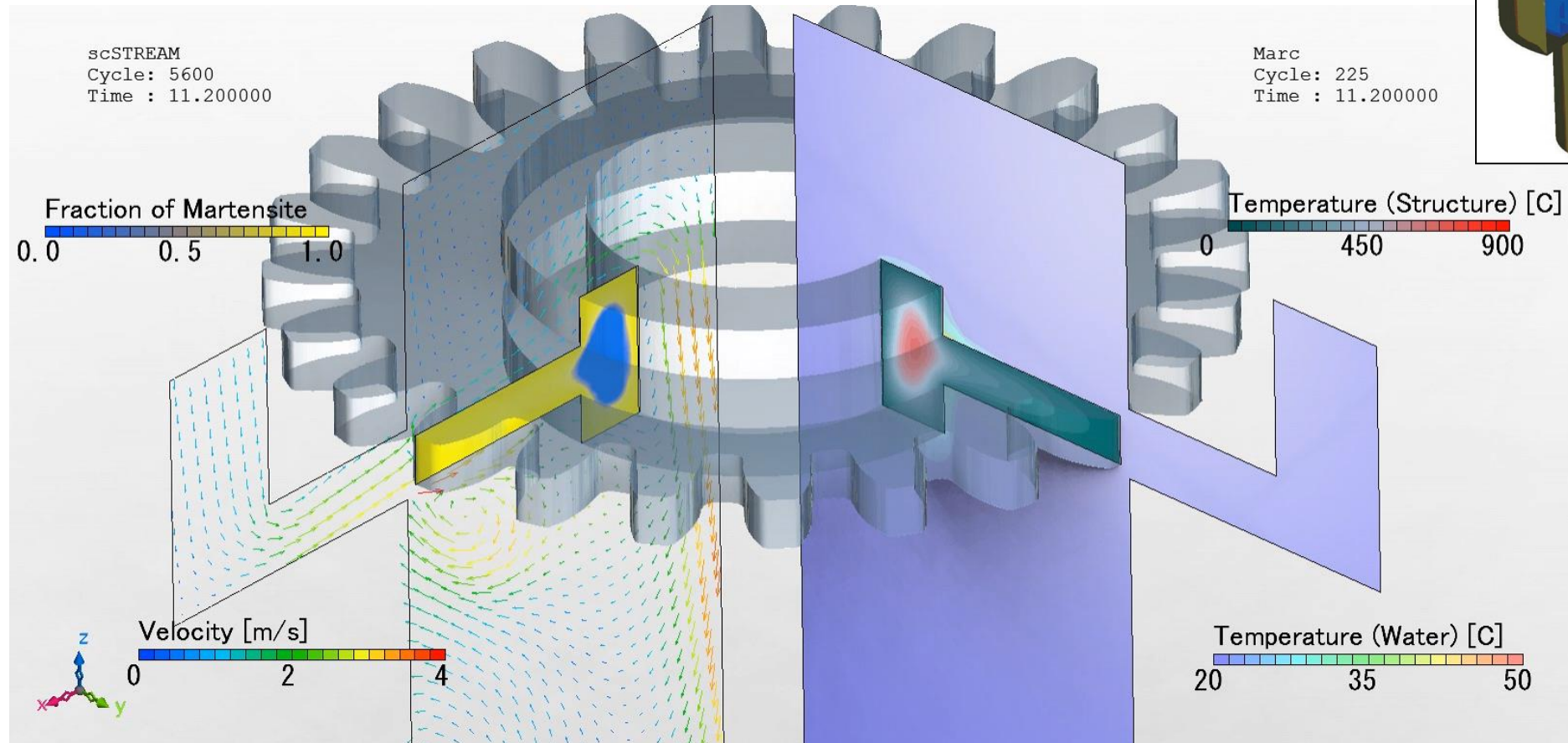
Adams – Marc: Windscreen Wiper



Adams – scFLOW: Platform of an Offshore Wind Turbine

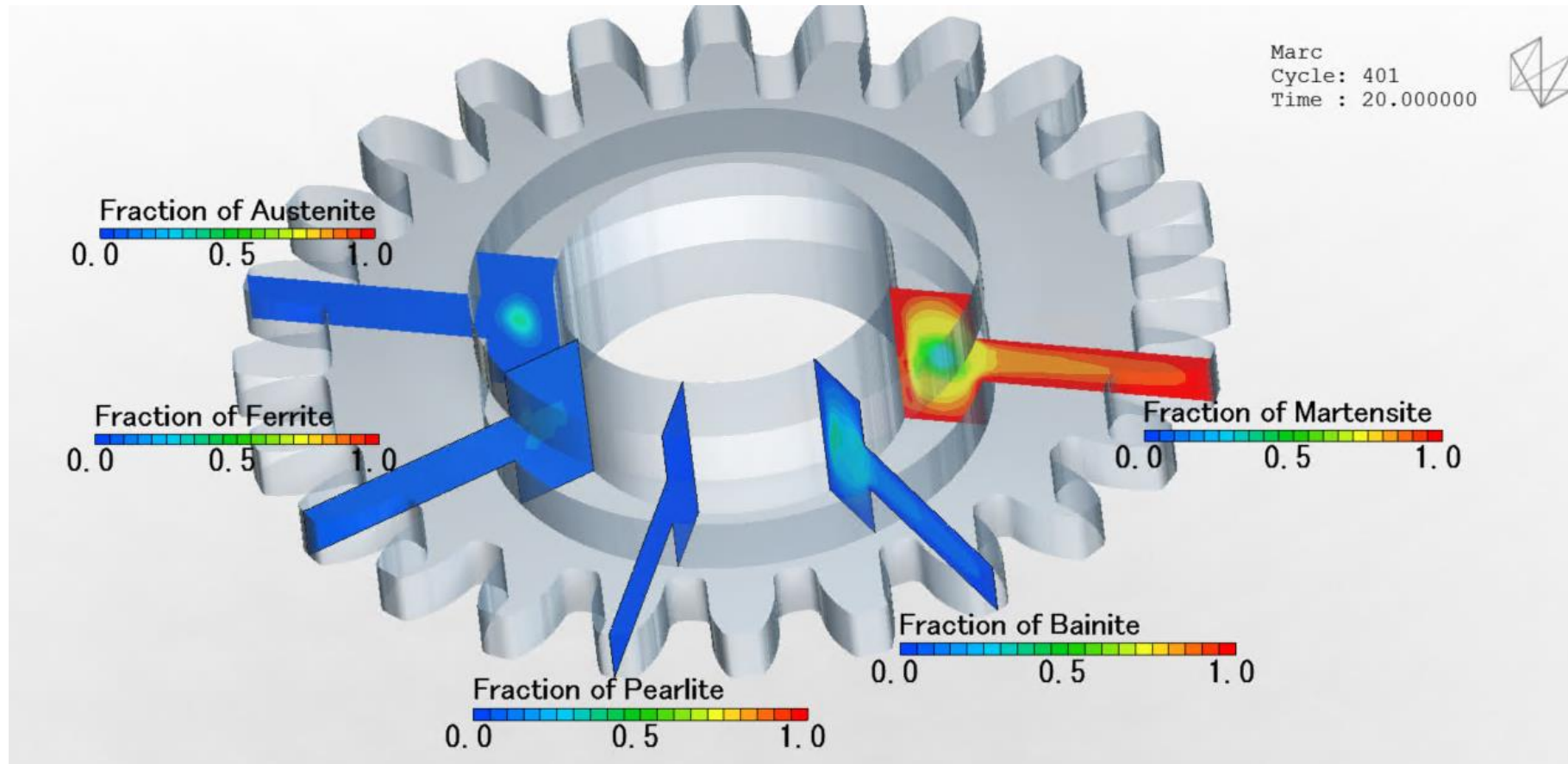


Marc – scSTREAM Heat transfer Coupling: Quenching of Gear

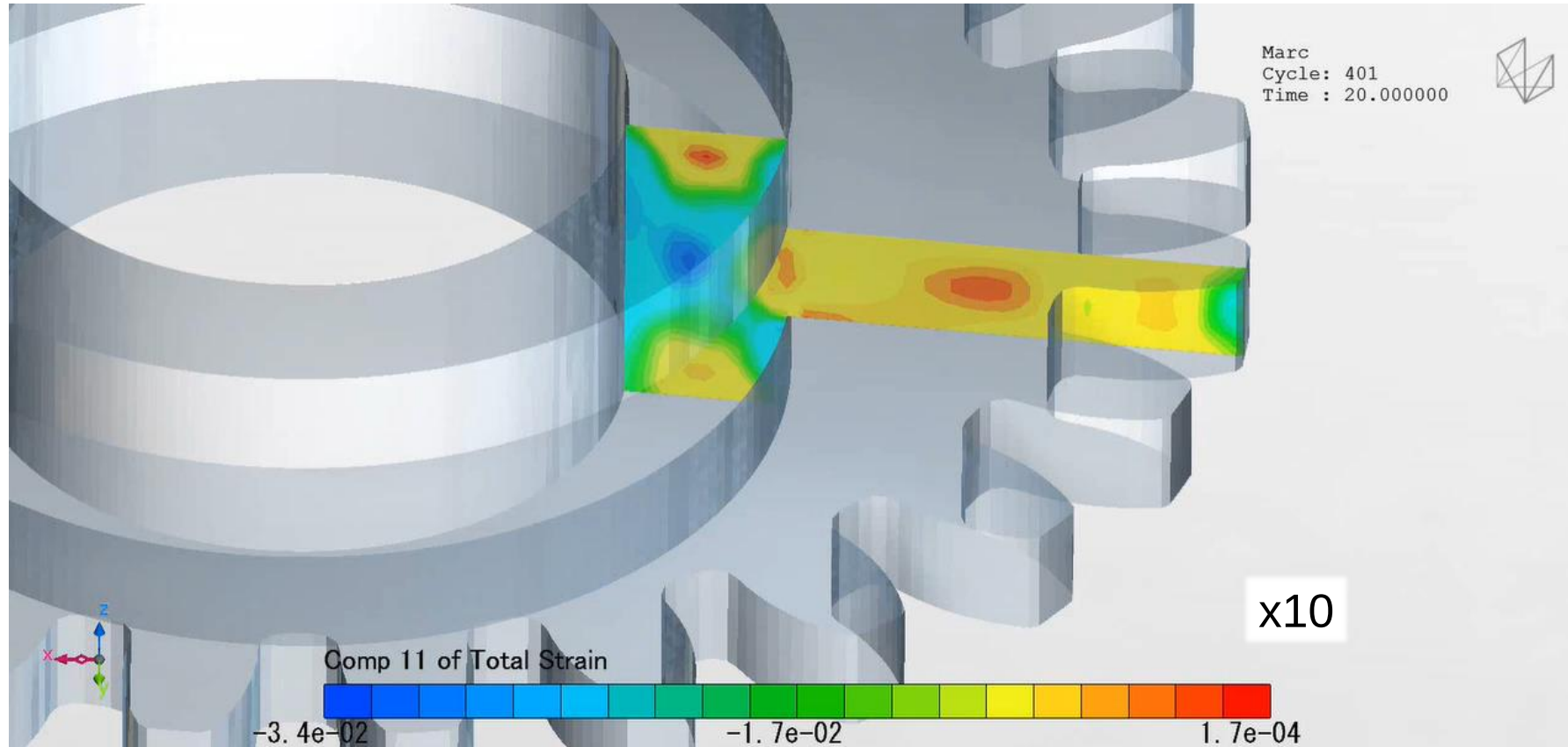


Marc – scSTREAM Heat transfer Coupling: Quenching of Gear

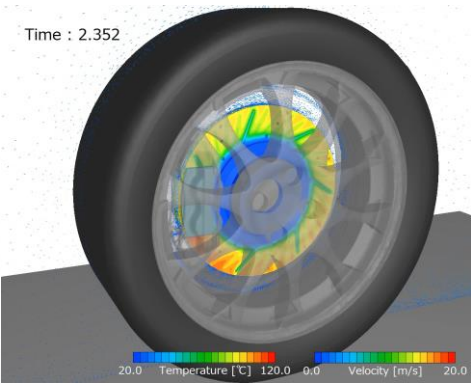
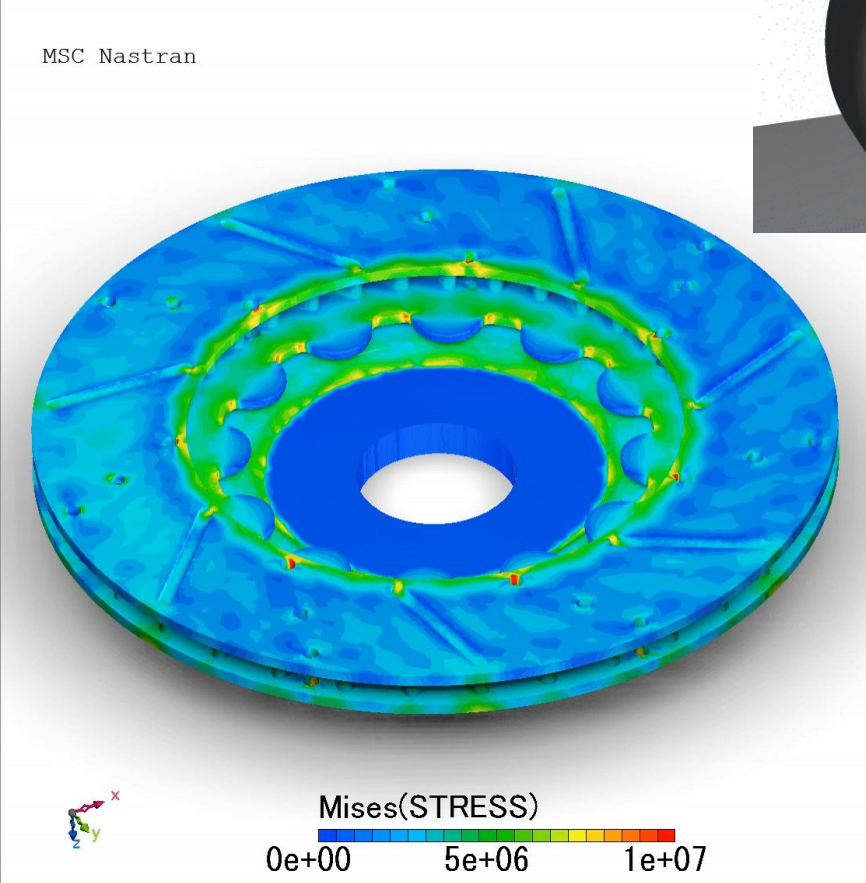
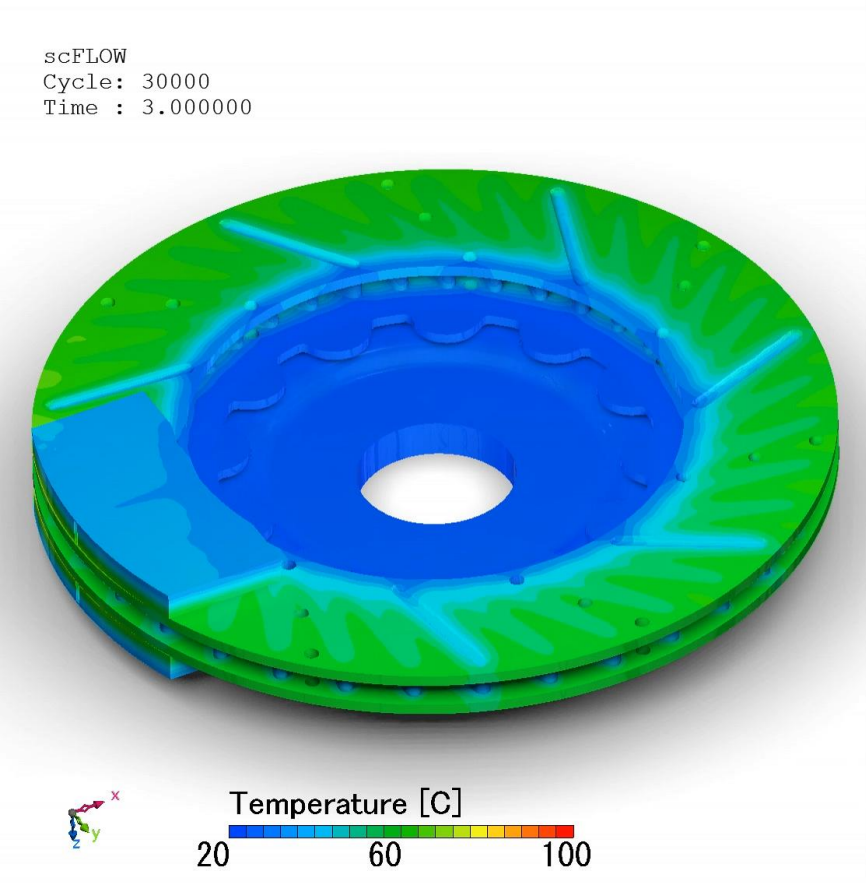
Material composition changes depending on the cooling rate



Marc – scSTREAM Heat transfer Coupling: Quenching of Gear

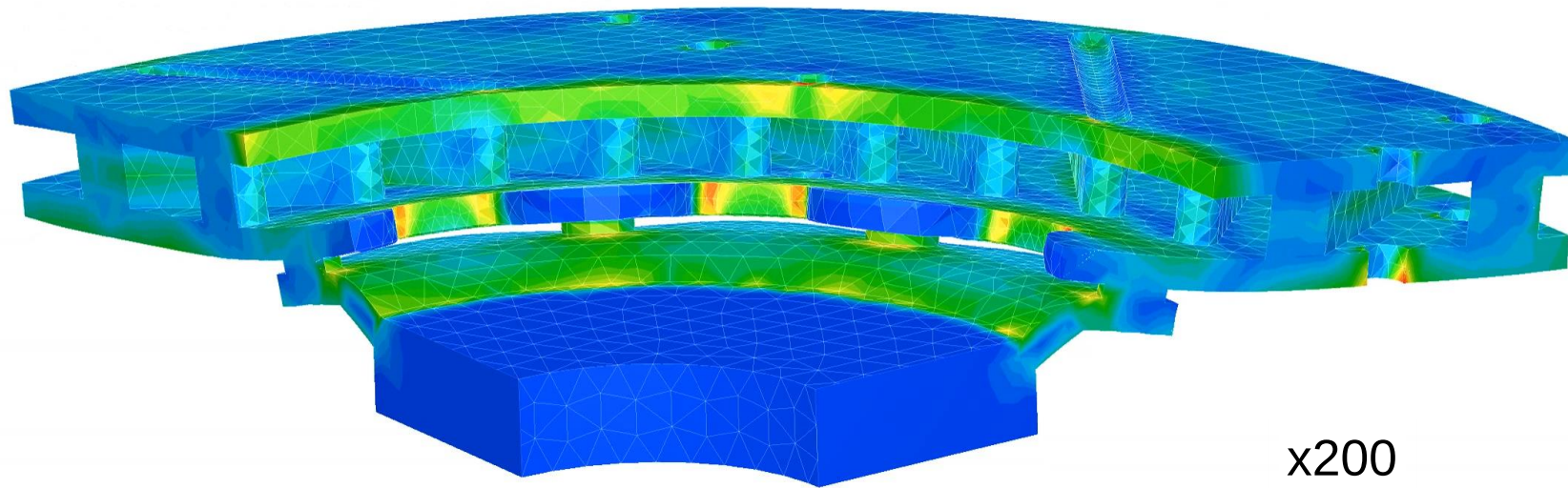


MSC Nastran – scFLOW: Disk brake thermal stress analysis



MSC Nastran – scFLOW: Disk brake thermal stress analysis

MSC Nastran Time : 2.750000

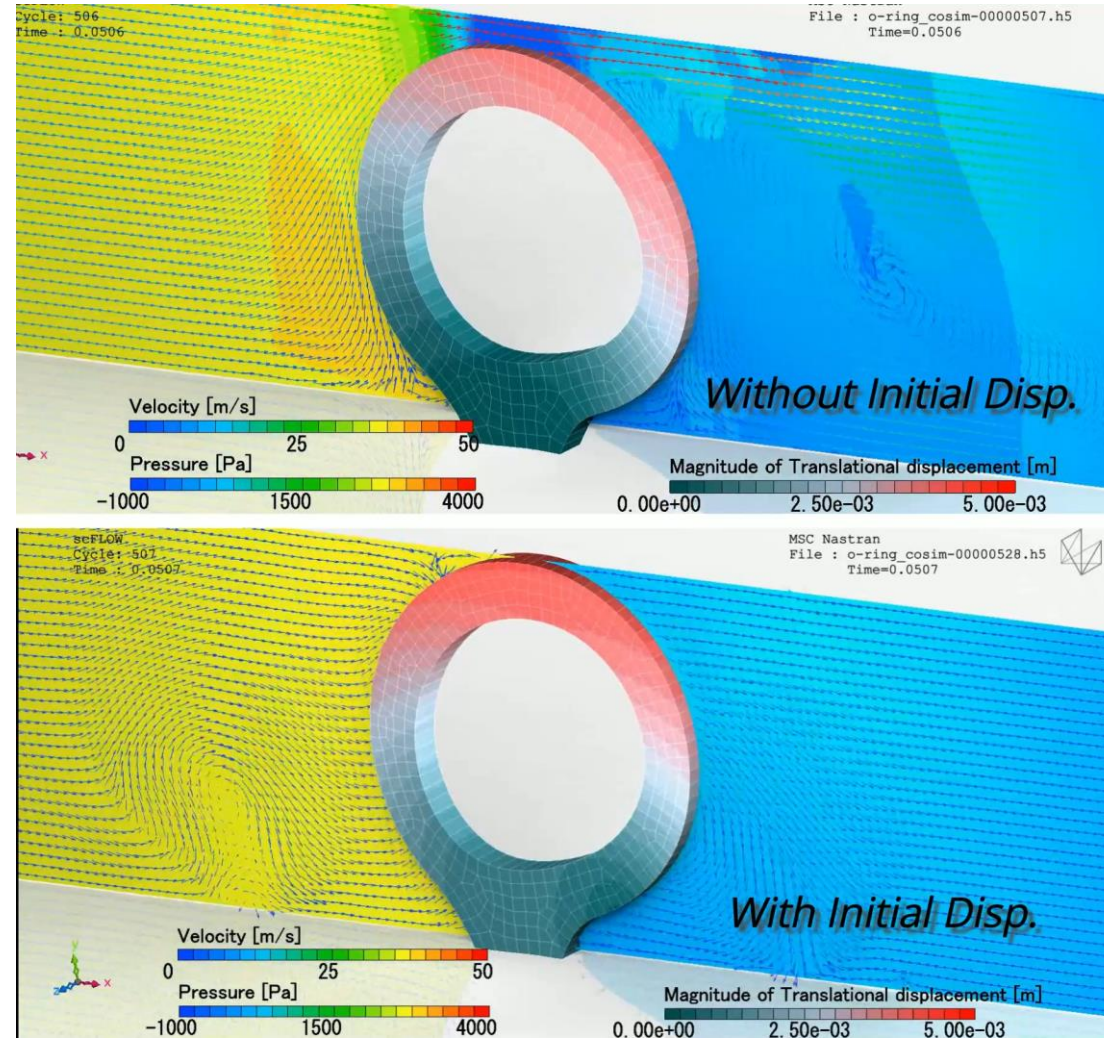


Mises(STRESS)

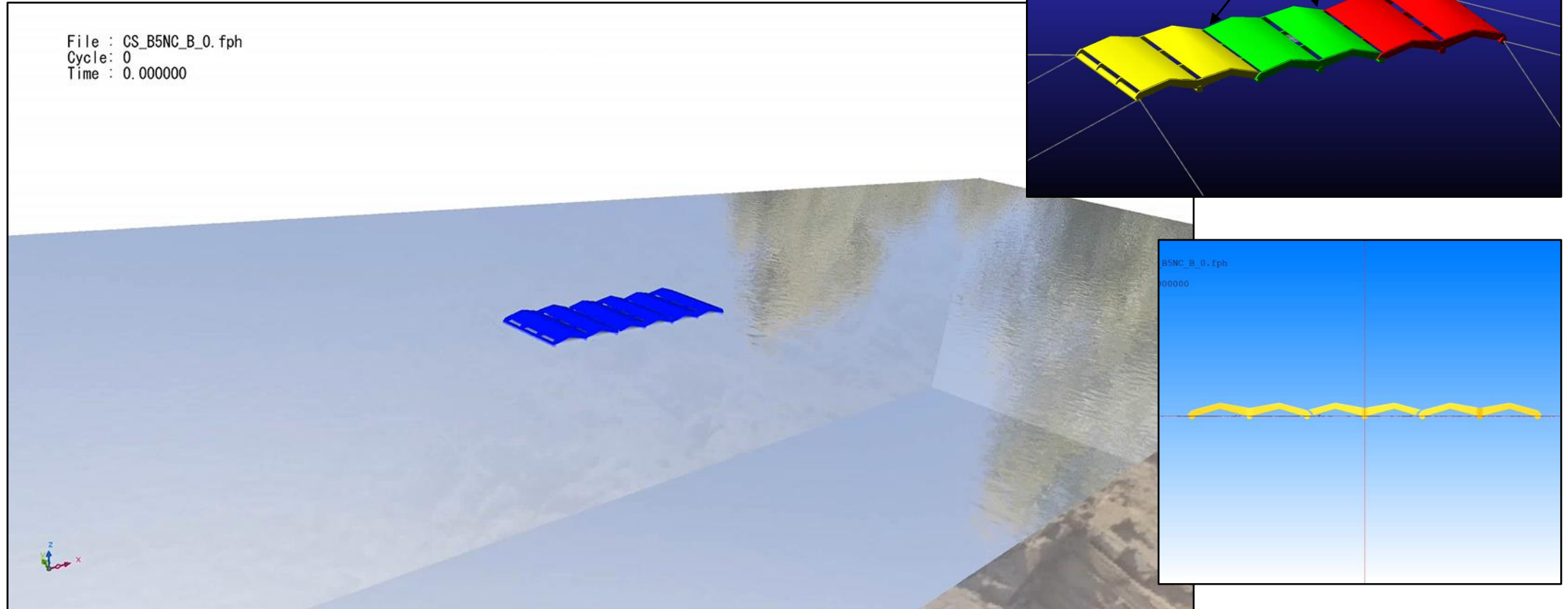


0e+00 5e+06 1e+07

MSC Nastran – scFLOW: Sealing rubber



Adams – scFLOW: Floating solar panels

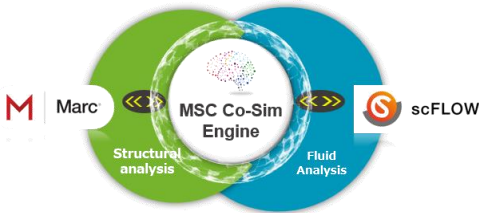
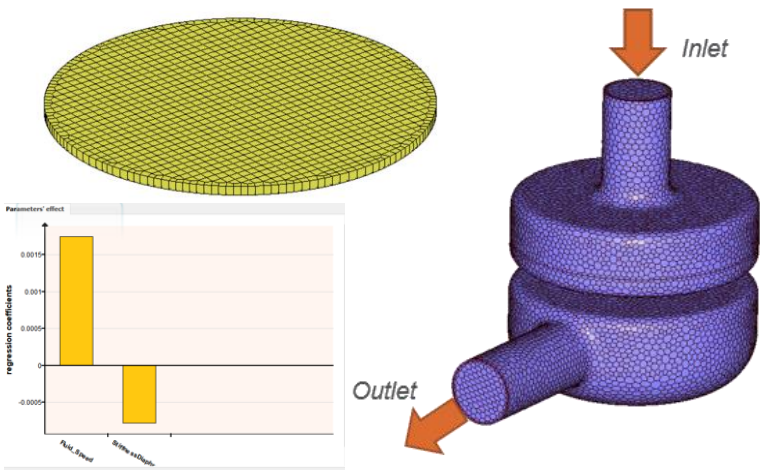


Pump membrane optimization with multi-physics interaction

Design of experiments

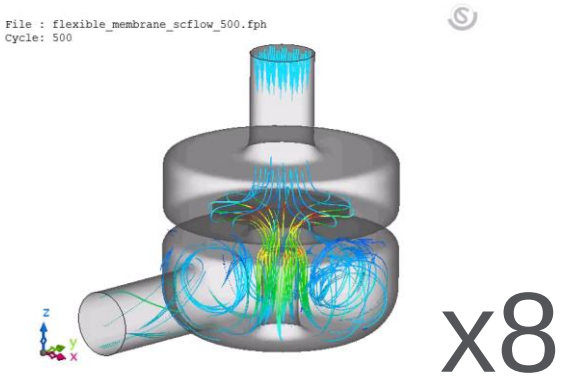
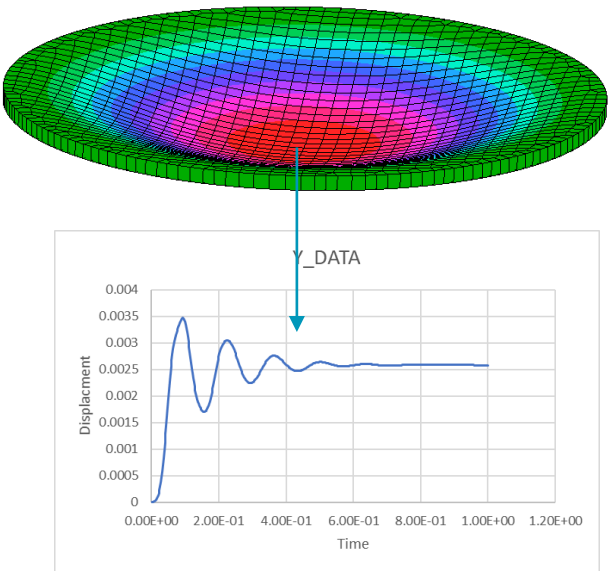
X: (2 variables)

1. Inlet Fluid Speed
2. Membrane Stiffness

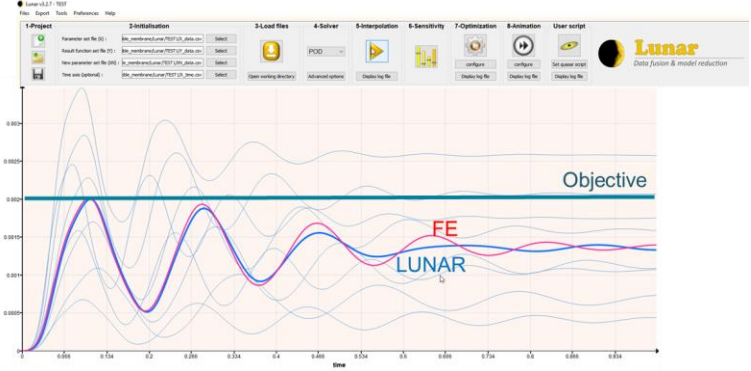


Y:

1. Displacement center of membrane

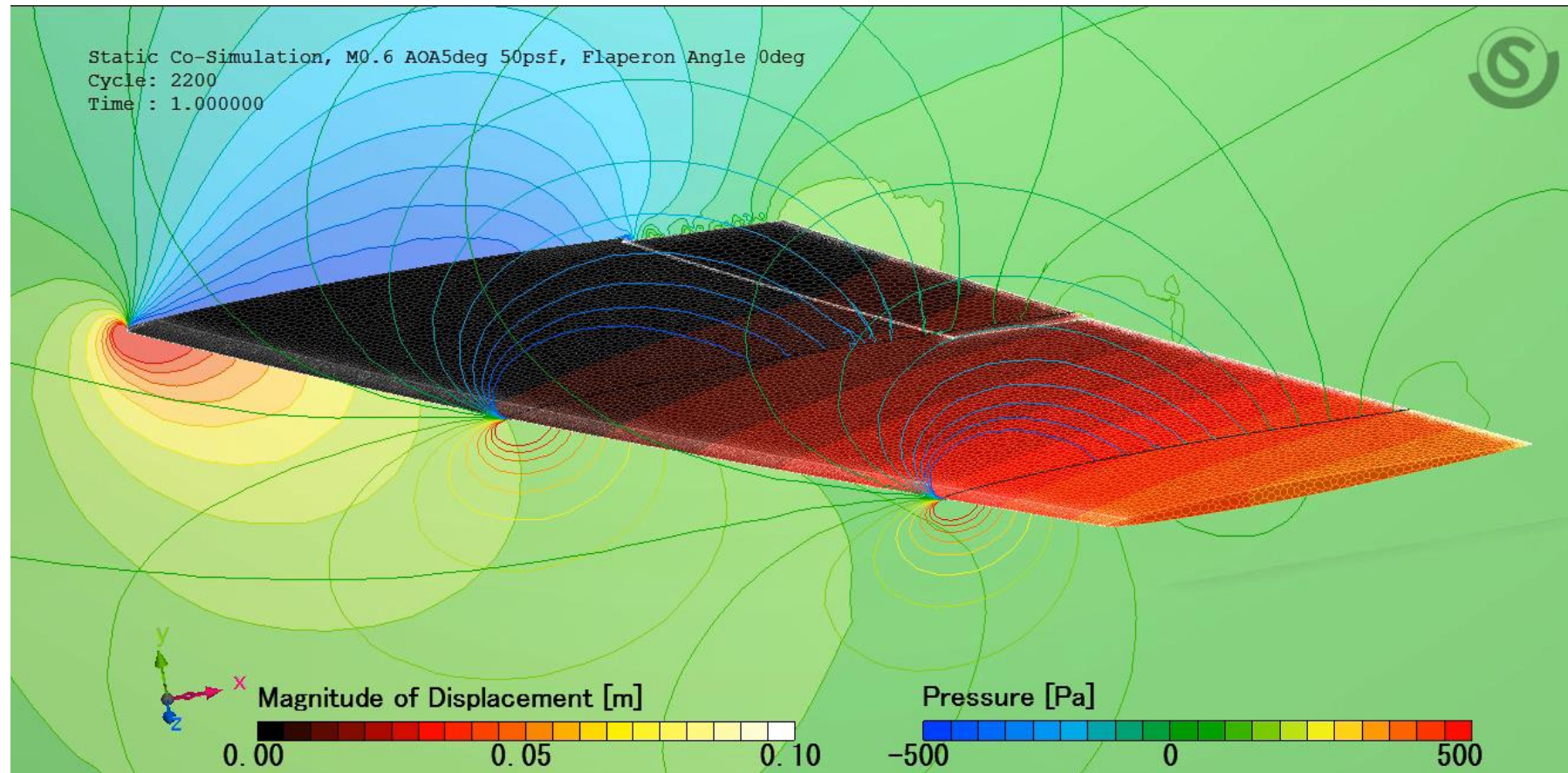


Result Verification



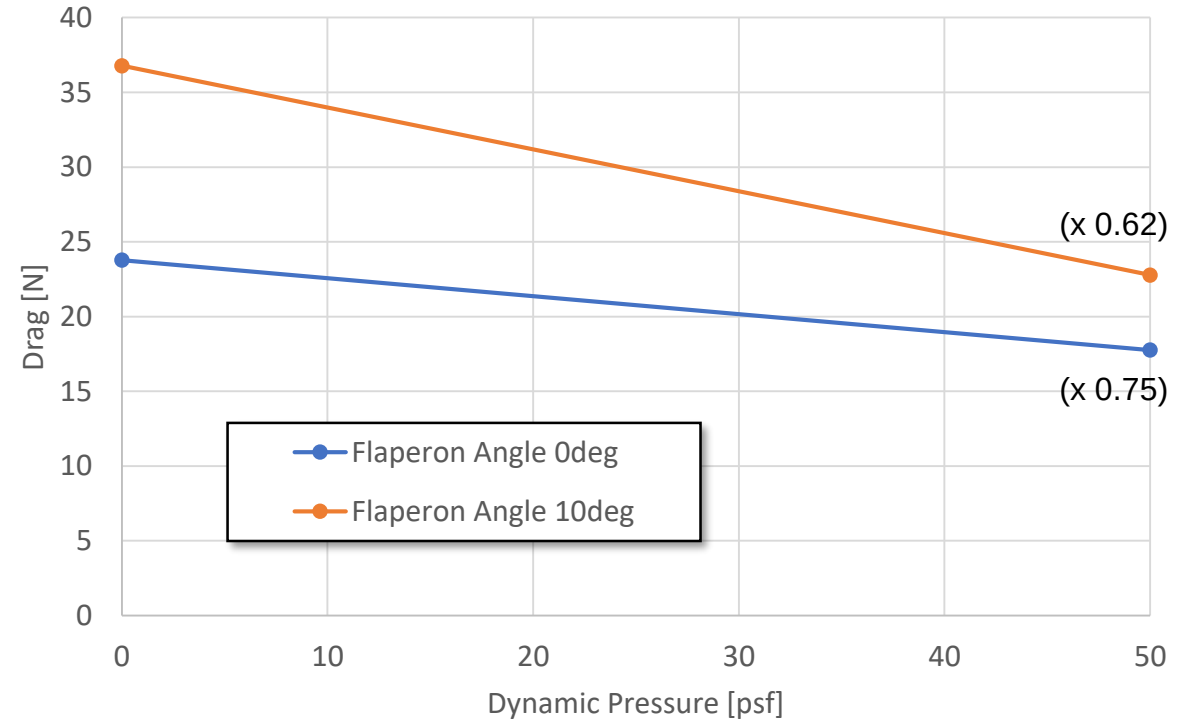
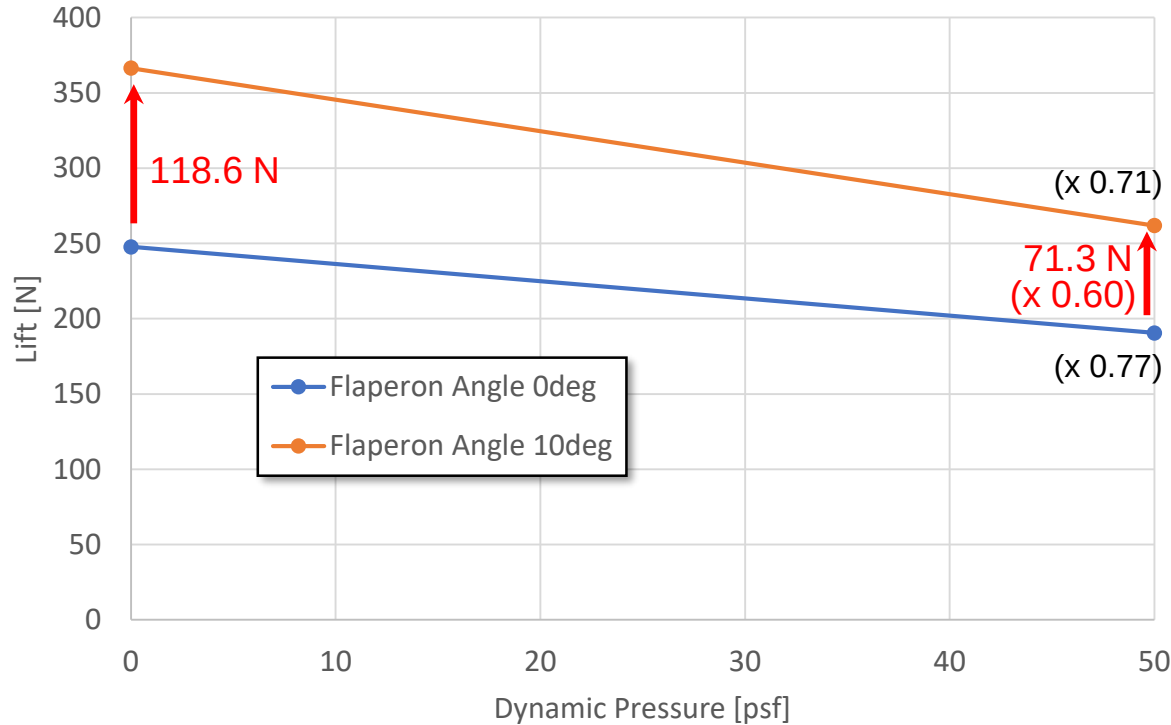
Steady flow – Static Nonlinear Structure Coupled Solution

Flaperon angle : 0 deg



Aeroelastic effect to Lift & Drag

- Lift & Drag gets smaller by aeroelastic effect
 - Aeroelastic factor is smaller for Flaperon angle 10 deg than 0 deg
 - Effective angle of Flaperon gets 0.6 times smaller by aeroelasticity



Hydrofoil scFLOW – Marc co- simulation

Challenge

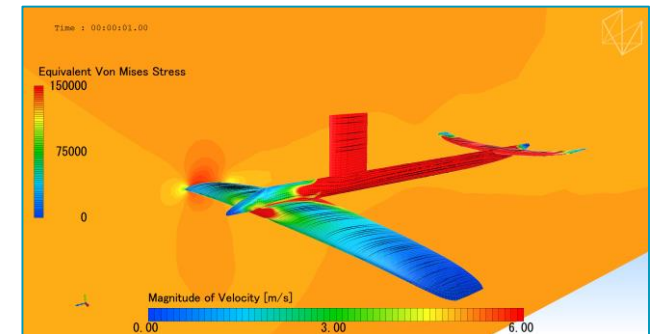
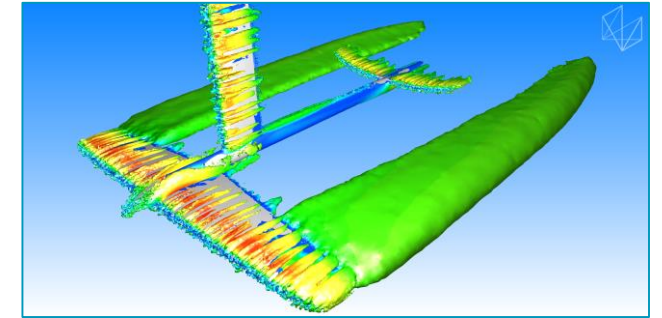
- Aeronautic is always on the edge of innovation, and concept designs need to be explored quickly and deeply
- The balance between hydrodynamic and structural resistance is very sharp, designers are targeting very high performances
- Involved phenomenon implies different physical domains, which can make simulations difficult to perform

Solution

- Digimat, Marc and scFLOW can deal individually with advanced composite material properties, structural behavior and fluid impact
- MSC CoSim can transfer CFD pressure loads (Cradle) to FEA calculation (Marc)
- scPOST help to review the multiphysics results at a glance

Benefits

- MSC One provides access to solutions covering the whole physical scenario
- Tools' connection allows a holistic consideration of the application
- Post-treatment capabilities can efficiently highlight innovations



Water meter piston simulation

Challenge

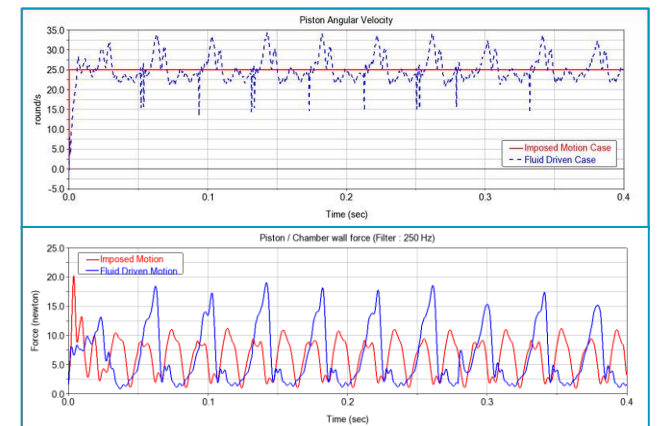
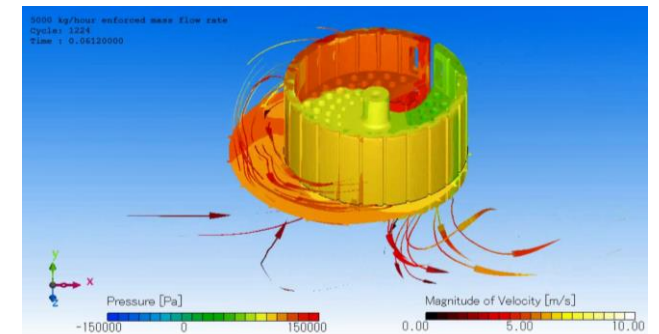
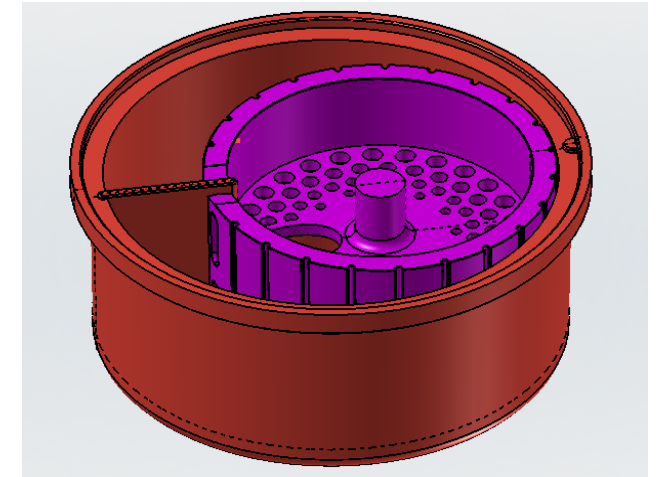
- The complex motion induced by the fluid flow must be finely captured, in order to capture the device working principle
- Contact between parts add further difficulty to the simulation
- For such mass-produced device, it's crucial to master wear areas to assure the best lifetime of the product

Solution

- scFLOW can deal with 6-DOF and motion combination of translation + rotation, and Adams can handle complex contact between parts
- MSC CoSim links both software tools
- scFLOW can also determine the forces acting on parts surfaces

Benefits

- Ability to consider the complex flow induced motion and the contacts
- Identification of wear areas
- Holistic consideration of the whole problematic



Announcing upcoming release!

MSC CoSim 2022.1

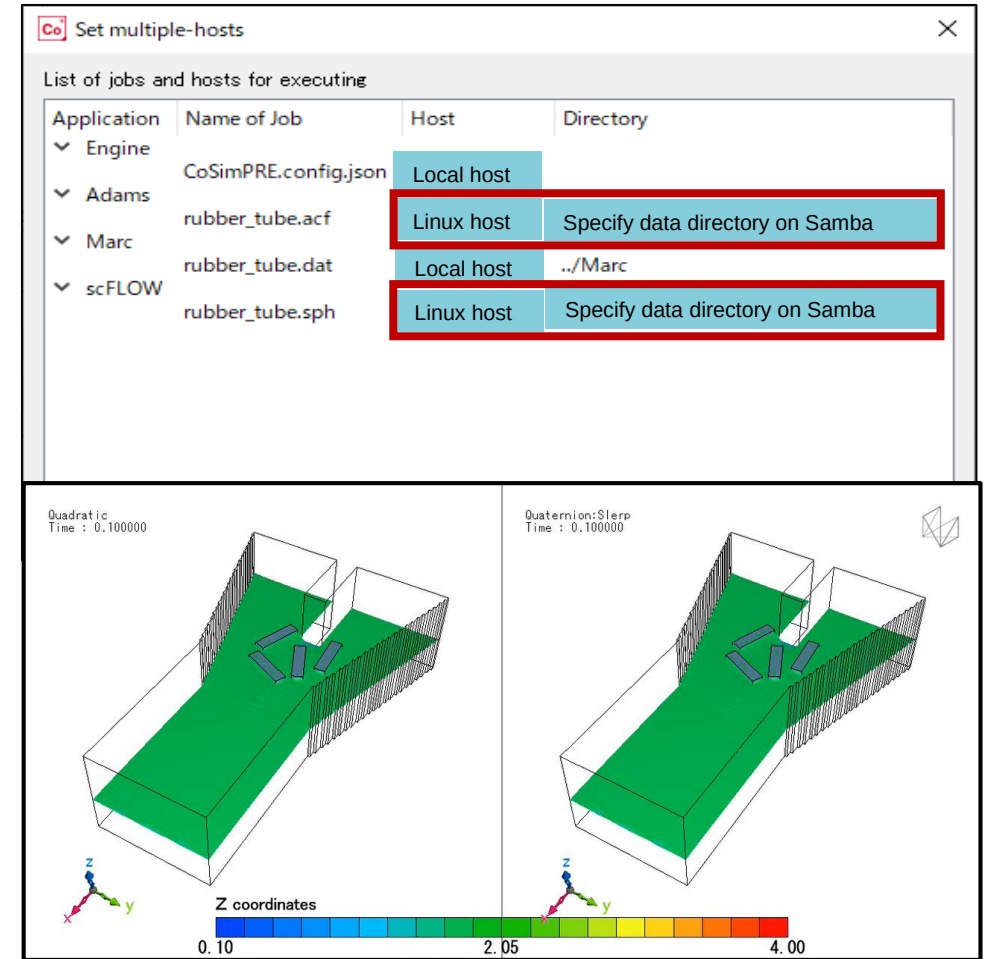
Sneak peek: MSC CoSim 2022.1

Oct 2022

- **NEW features addressing**
 - Usability
 - Stability
 - Accuracy
- **Updated Demo Guide**

New features

- Multiple host support
- Improved treatment of large deformations of solid structures in CFD computations
- Enhancement of time interpolation for Adams - scFLOW and Marc - Adams
- More accurate pressure mapping in Marc - scFLOW
- Improved algorithm for computation of follower forces in moving/deforming bodies
- Various improvements of Pre-processor



Thank You!